Your Forces Worksheet, Decoded

01

The class worksheet "Forces (including circular motion)", mapped to what you've already practiced here, plus the pieces that were missing. Those are on pages 2–4.

Coverage map: worksheet problem types → where to practice

WORKSHEET PROBLEM TYPE	WHERE ON THE SITE	STATUS
Net force zero or not? (projectile, constant speed on a curve, Moon)	Newton's Laws Quest · Circular Quest	✓ covered
Reaction-force pairs · scale readings over pulleys	Newton's Laws · Tension Quest · Quizzes	✓ covered
Elevator tension · weight on the Moon / in space	Systems & Gravity Quest	✓ covered
Helicopter lifting / lowering a load (2 bodies, vertical)	**Page 2** of this sheet	★ new
Boxes pulled in a line · Atwood · table + hanging mass	Tension & Pulleys Quest	✓ covered
3 blocks (hanging–table–hanging) **with friction** · incline + hanging **with friction**	**Page 2**: stack-and-add + friction direction	★ new
Slanted push + friction (N ≠ mg) · symmetric strings at 53°	Friction Quest · Tension Quest · **page 4** (37°/53° trick)	★ added
Block resting on an incline: friction = ? · pushed at constant speed	**Page 4**: "static friction is only what's needed"	★ new
Ramp + kinematics (how long, how fast, how far up and back)	Friction Quest boss · **page 4** pipeline	★ added
Cup of water in a vertical circle · vine/rope at the bottom	Circular Quest · Predict & Reveal	✓ covered
Ferris wheel (rev/min → speed; seat force at top AND bottom)	**Page 3**	★ new
Puck on a table, string through a hole to a hanging mass	**Page 3**	★ new
"Does it slide off / does the vine snap?" (needed vs available)	**Page 4**	★ new

Read the worked solutions like a pro: notation decoder

$F_{net} = ma$	Newton's 2nd law. Write one per object.
F_{fr}, F_{sr}, f	Friction (handwriting varies). $F_{fr} = \mu F_N$ when sliding.
F_g, w, mg	Weight. With g = 10, a 9-kg block → 90 N.
a_c	Centripetal acceleration v^2/r, toward the center
+) under equations	"Add these equations": tensions cancel (page 2)
a↓ / a↑ beside an FBD	The direction chosen as + for that object
★ / *	Final answer

g = 10 vs g = 9.8

Your class key uses **g = 10 m/s²** (easier arithmetic). This site mostly uses 9.8, so answers can differ by about 2%. That's not a mistake.

- On class tests: use whatever your teacher uses (looks like **10**).
- On the AP exam: 9.8 or 10 both earn credit if you're consistent.
- Quick convert: answer with 9.8 ≈ answer with 10 × 0.98.

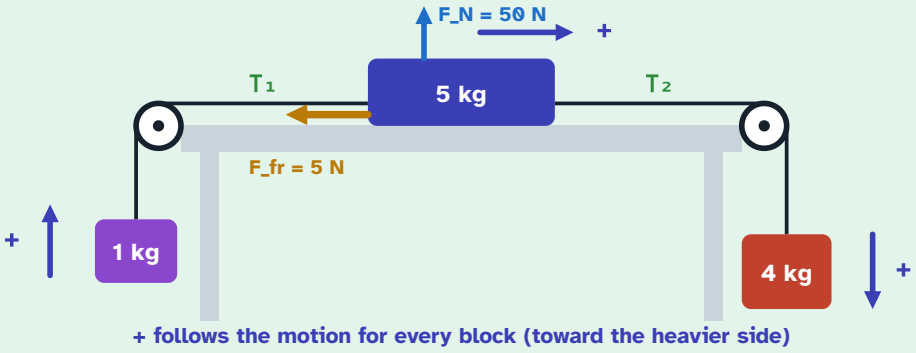
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Weight: m × g → 60 kg × 10 = 600 N
```

The Stack-and-Add Method

The key solves every multi-object problem the same way: one $F_{net} = ma$ per object, then add the equations. Internal tensions cancel like magic.

02

Example: hanging – table – hanging, with friction

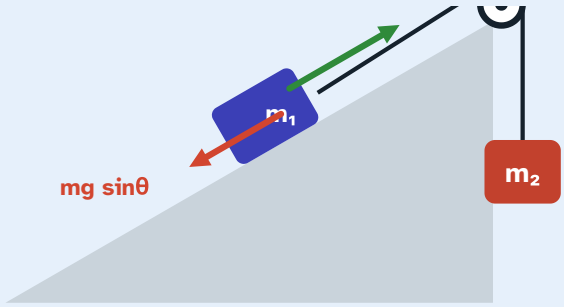


1. **Pick the direction of motion** first. The heavier side wins: 4 kg (40 N) beats 1 kg (10 N), so the system moves **toward the 4 kg**.
2. **Make + point along the motion for EVERY object**: up for the 1 kg, right for the 5 kg, down for the 4 kg. The string "bends" the axis around the pulleys.
3. **Friction opposes the motion**, so on the 5 kg it points left: $F_{fr} = \mu F_N = 0.1 \times 50 = 5$ N.
4. **Write $F_{net} = ma$ for each object, then ADD.** T_1 and T_2 each appear once as + and once as –, so they cancel.
5. **Plug a back in** to any single equation to get each tension.

```
1 kg:  T₁ - 10 = 1·a
5 kg:  T₂ - T₁ - 5 = 5·a
4 kg:  40 - T₂ = 4·a
─────────────────────────
+)     25 = 10·a → a = 2.5 m/s²
T₁ = 10 + 1(2.5) = 12.5 N
T₂ = 40 - 4(2.5) = 30 N
```

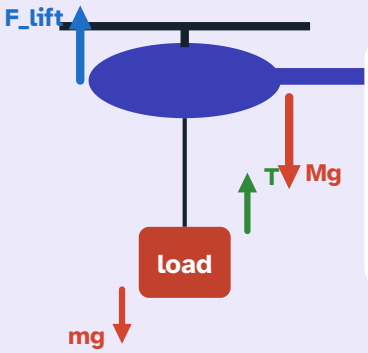
Shortcut check: a = (driving forces – resisting forces) ÷ (total mass) = (40 – 10 – 5) ÷ 10 = 2.5 m/s². Same answer, and only one line.

Same trick on a ramp



- Along the ramp, gravity's pull is **mg sinθ**. Into the ramp it's **mg cosθ**, so $F_N = mg\cos\theta$.
- Compare the hanging weight with mg sinθ to find which way it moves. Friction then points **the other way**.
- With friction: driving = hanging weight – mg sinθ – μmg cosθ.

Vertical version: helicopter + load




```
system:
F_lift-(M+m)g
=(M+m)a

load:
T - mg = ma
```

- **Whole system** (heli + load): $F_{lift} - (M+m)g = (M+m)a$ → the lift force.
- **Load alone**: T – mg = ma → the cable tension.
- Moving **down** and speeding up? Then a points down, so the lift is *less* than the total weight.

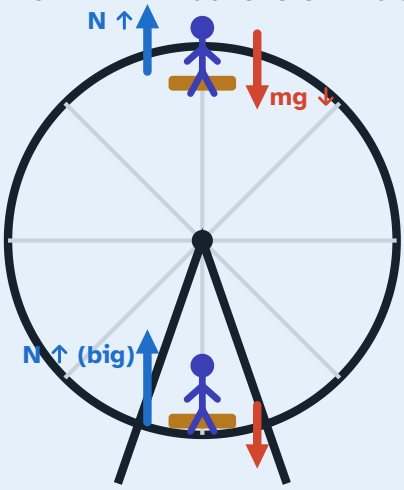
Two Circular Problems We Hadn't Covered

03

The Ferris wheel, and the puck tied through a hole to a hanging mass. Both are classic test favorites.

Ferris wheel vs. coaster loop: is the seat UNDER you or OVER you?

FERRIS WHEEL · rider sits ON the seat

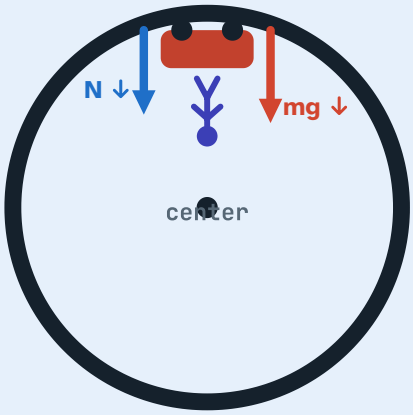


Ferris wheel, top: you sit ON the seat, so the seat pushes **up** (away from the center). Gravity points down (toward the center).

$$mg - N = mv^2/r \rightarrow N = m(g - a_c)$$

You feel lighter. Bottom: N – mg = mv^2/r → you feel heavier.

COASTER LOOP · rider upside down INSIDE



Coaster loop, top: you're upside down INSIDE the track, so the track pushes **down** (toward the center).

$$mg + N = mv^2/r$$

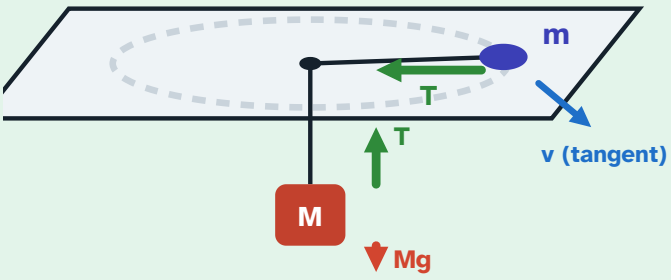
Same position, opposite N! **Always draw where the surface touches you** before writing the equation.

"Turns 4 times a minute": the rpm pipeline

rev/min	period T	speed v	a_c	force
4 rev per min	60 s ÷ 4 = 15 s	v = 2πr / T	a = v² / r	F_net = m·a_c

Puck on a table, string through a hole, hanging mass

top: $T = mv^2/r$
bottom: $T = Mg$ (at rest)



1. **Hanging mass M is at rest** (a = 0): T = Mg.
2. **The same string pulls the puck m** toward the hole. That tension IS the centripetal force: $T = mv^2/r$.
3. **Chain them:** $Mg = mv^2/r \rightarrow v = \sqrt{(Mgr / m)}$.

Intuition: spin the puck faster and it needs more pull than M's weight can give. M gets yanked **up** and the circle widens until things balance again.

$$Mg = T = mv^2/r$$

Needed vs. Available + Your Angle Toolbox

"Does it slide off? Does the vine snap?" questions, the 37°/53° shortcut, and the friction fact that trips everyone up.

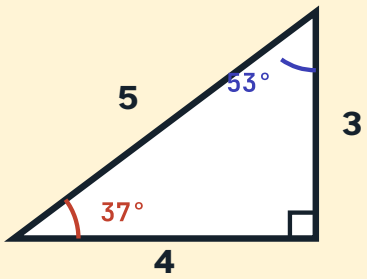
04

Threshold problems: compare what the circle NEEDS with what the force CAN give

SITUATION	NEEDED (TOWARD THE CENTER)	AVAILABLE (THE LIMIT)	IF NEEDED > AVAILABLE…
Box on a spinning merry-go-round / coin on a turntable	mv^2/r	Max static friction $\mu_s F_N = \mu_s mg$	It slides off. At the edge: $v = \sqrt{(\mu_s gr)}$
Swinging on a vine/rope, at the bottom	$T = mg + mv^2/r$	Breaking strength	The vine snaps. Justify with numbers!
Cup or bucket upside down at the top	mv^2/r	At least mg (gravity alone)	If $mv^2/r < mg$, the water falls out. Minimum $v = \sqrt{(gr)}$
Car over a hilltop	mv^2/r	At most mg (N can't pull down)	The car leaves the road at $v > \sqrt{(gr)}$

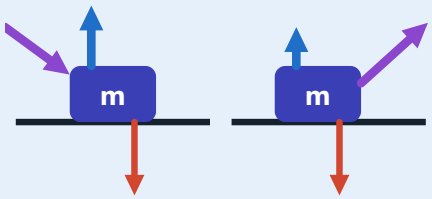
FRQ move: "The tension needed is 1236 N, which is more than the 1000 N breaking strength, so the vine breaks." A number on each side plus a comparison word scores the point.

The 3-4-5 triangle



sin 37° = cos 53° = 0.6
cos 37° = sin 53° = 0.8
Problems using 37° or 53° are inviting this shortcut, so no calculator is needed.

Slanted force → N changes



N = mg + F sinθ — **push ↘: N bigger**

N = mg − F sinθ — **pull ↗: N smaller**

Push down at θ: N = mg + F sinθ (more friction).
Pull up at θ: N = mg − F sinθ (less friction).
Only F cosθ moves the box forward.

Static friction = only what's needed

A block **at rest** on a 30° ramp: friction = **mg sinθ** (whatever balances it), **not** $\mu_s F_N$.

$\mu_s F_N$ is the **maximum**, used only when it's "just about to slip".

Pushed at **constant speed**? a = 0, so friction = your push (70 N push → 70 N friction).

Ramp + kinematics: forces give a, then a gives motion

1. F_{net} along the ramp → a (frictionless: a = g sinθ; sliding up with friction: a = −(g sinθ + μg cosθ)). **2.** Pick a kinematics equation: $\Delta x = v_0 t + \frac{1}{2}at^2$ · $v = v_0 + at$ · $v^2 = v_0^2 + 2a\Delta x$. **3.** "How far up?" means v = 0 at the top. **Frictionless round trip:** the time up = the time down, so total = 2 × t_{up}, and it comes back at the speed it started with.

10 New Practice Problems

Same types as the class worksheet, but new numbers. Use **g = 10 m/s²**.
Draw an FBD for every one. Cover the answers until you're done!

05

P1 · Ferris wheel

A 50-kg rider is on a Ferris wheel of radius 12 m that turns 3 times per minute. (a) Find the period and the speed. (b) Find a_c. (c) Find the seat's force on the rider at the bottom and at the top (magnitude and direction). (d) Challenge: how many rev/min would make the rider feel weightless at the top?

P2 · Puck + hole

A 0.20-kg puck circles on a frictionless table at radius 0.80 m. Its string passes through a hole to a hanging 0.50-kg mass that stays at rest. (a) Find the tension. (b) Find the puck's speed. (c) To double the speed at the same radius, what hanging mass is needed?

P3 · Will it snap?

A 60-kg climber swings on a 6.0-m rope and moves at 5.0 m/s at the lowest point. The rope is rated for 900 N. (a) Does it hold? Justify. (b) What's the fastest safe speed at the bottom?

P4 · Slide off?

A coin sits 0.15 m from the center of a turntable, with $\mu_s = 0.40$. (a) What's the maximum speed before it slides? (b) How many rev/min is that?

P5 · 3 blocks + friction

A 2-kg block hangs on the left and a 4-kg block hangs on the right, joined by strings over pulleys to a 6-kg block on a table ($\mu_k = 0.20$). Find the acceleration and both tensions.

P6 · Ramp + hanging

A 5-kg block on a 37° ramp ($\mu_k = 0.25$) is tied over a pulley at the top of the ramp to a hanging 6-kg block. Which way does it move? Find a and T.

P7 · Slanted pull

You pull a 10-kg crate with 50 N at 37° above the horizontal. $\mu_k = 0.30$. Find the normal force and the acceleration.

P8 · Resting on a ramp

An 8-kg box sits at rest on a 37° ramp with $\mu_s = 0.90$. What is the friction force on it? (Careful!)

P9 · Ramp kinematics

A block is released from rest at the top of a frictionless 30° ramp 5.0 m long. How long does it take to reach the bottom, and how fast is it going?

P10 · Helicopter

A 3000-kg helicopter lifts a 500-kg crate on a cable, accelerating upward at 1.0 m/s². Find the lift force and the cable tension.

Answers (cover these until you've tried!)

1. T = 20 s, v = 2π(12)/20 = 3.77 m/s; a_c = 1.18 m/s²; bottom N = 50(10 + 1.18) = 559 N ↑, top N = 50(10 − 1.18) = 441 N ↑ (still up!); weightless when v = √(gr) = 10.95 m/s → T = 6.9 s → about 8.7 rev/min.

2. T = Mg = 5.0 N; v = √(Tr/m) = √(5 × 0.8 / 0.2) = 4.47 m/s; 2v needs 4× the force → M = 2.0 kg.

3. T = mg + mv²/r = 600 + 60(25)/6 = 850 N < 900 N, so it holds. Max: 900 − 600 = 60v²/6 → v = 5.48 m/s.

4. v = √($\mu_s$gr) = √(0.4 × 10 × 0.15) = 0.775 m/s; f = v/(2πr) = 0.82 rev/s ≈ 49 rev/min.

5. It moves toward the 4 kg. F_{fr} = 0.2 × 60 = 12 N; a = (40 − 20 − 12)/12 = 0.667 m/s²; T_{left} = 20 + 2a = 21.3 N; T_{right} = 40 − 4a = 37.3 N.

6. 60 N > mg sin37° = 30 N, so the hanging block goes down and the ramp block goes up. F_{fr} = 0.25 × 50 × 0.8 = 10 N (down the ramp). a = (60 − 30 − 10)/11 = 1.82 m/s²; T = 60 − 6a = 49.1 N.

7. N = 100 − 50(0.6) = 70 N; F_{fr} = 0.3 × 70 = 21 N; a = (50(0.8) − 21)/10 = 1.9 m/s².

8. At rest → friction = mg sinθ = 80 × 0.6 = **48 N** (up the ramp). NOT μ_sN = 57.6 N; that's the maximum it could give.

9. a = g sin30° = 5 m/s²; 5 = ½(5)t² → t = 1.41 s; v = at = 7.07 m/s.

10. System: F_{lift} = (3500)(10 + 1) = 38,500 N. Crate: T = 500(10 + 1) = 5,500 N.